

Random Password Generator

Introduction

The Random Password Generator is a Python script that creates secure, randomized passwords. It allows users to specify the desired length and complexity, ensuring strong security for online accounts.

It uses python fundamentals concepts like if statement, user input.

Features

1. Generate passwords of customizable length.
2. It includes uppercase letters, numbers, and special characters.
3. Ensures strong, unpredictable passwords.
4. Simple and easy to use.

Code Explanation

Dependencies

This script uses Python's built-in *random* and *string* modules, so no additional dependencies are required.

Importing Required Libraries

```
import random
import string
```

User Input Handling

```
length = int(input("Enter password length: "))
```

Character Selection

```
use_upper = input("Include uppercase letters? (y/n): ").lower() == 'y'
use_lower = input("Include lowercase letters? (y/n): ").lower() == 'y'
use_digits = input("Include numbers? (y/n): ").lower() == 'y'
```

Building Character Sequence

```
char_seq = ""
if use_upper:
    char_seq += string.ascii_uppercase
```

```
if use_lower:
    char_seq = ""
if use_upper:
    char_seq += string.ascii_uppercase
if use_lower:
    char_seq += string.ascii_lowercase
if use_digits:
    char_seq += string.digits
if use_symbols:
    char_seq += "!@#$%^&*?/"
```

Validation

```
if not char_seq:
    print("You must select at least one character type!")
    return
```

Generating Password :

```
password = ''.join(random.choice(char_seq) for _ in range(length))
```

Password Output

```
print(f"Generated Password: {password}")
```

Save the code

Save the code with name followed by **.py** extension
random_pass.py

Run the code

random_pass.py

Example Output

```
PS E:\python> & C:/Users/KHATRI/AppData/Local/Programs/Python/Python314/python.exe
e:/python/random_pass.py
Enter password length: 12
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include numbers? (y/n): y
Include special characters? (y/n): y
```

Generated Password: ?FPN%nhDz@u0

```
PS E:\python> & C:/Users/KHATRI/AppData/Local/Programs/Python/Python314/python.exe  
e:/python/random_pass.py  
Enter password length: 21  
Include uppercase letters? (y/n): y  
Include lowercase letters? (y/n): n  
Include numbers? (y/n): y  
Include special characters? (y/n): n  
Generated Password: QPKCV3TGP6NBVKTILF12  
PS E:\python>
```

Possible Enhancements

GUI with Tkinter for better user experience.